device.py

`src/moe_pruning/utils/device.py · 83 lines · module moe_pruning.utils.device`

What it does

Device and dtype selection.

How it works

- `resolve_device()`: `auto` picks CUDA, then MPS, then CPU; an explicitly requested device that is unavailable is an error, never a silent fallback.
- `resolve_dtype()` maps `float32` / `float16` / `bfloat16`; `check_dtype_supported()` warns about poor combinations.
- `synchronize()` and `empty_cache()` wrap the CUDA/MPS specifics.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`class DeviceUnavailableError`	Raised when an explicitly requested device is not available.
`is_available(kind)`	True if a device type (cpu / cuda / mps) can be used.
`resolve_device(requested)`	Resolve a device request.
`resolve_dtype(name)`	Map MODEL_DTYPE to a torch dtype.
`check_dtype_supported(device, dtype)`	Warn about poor device/dtype combinations.
`log_device(device, dtype)`	Log the selected device and dtype.
`synchronize(device)`	Wait for pending GPU work.
`empty_cache(device)`	Release cached GPU memory.

Connections

Used by: `scripts/convert_model.py`, `scripts/evaluate.py`, `scripts/export_model.py`, `scripts/train_biases.py`, `src/moe_pruning/model/loader.py`, `src/moe_pruning/model/placement.py`, `src/moe_pruning/serialization/checkpoint.py`